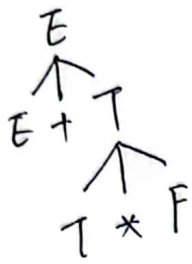


课后习题(4) - 自底向上的语法分析.

于梦洁
3021244141

1. 解: $E \Rightarrow E+T \Rightarrow E+T * F$ 因此 $E+T * F$ 是文法的一个句型.

语法树



因此短语有 $E+T * F$, $T * F$ 直接短语有 $T * F$, 句柄为 $T * F$

2. 解: (1) $(((a, a), \wedge, (a)), a)$ 的最右推导为

$$\begin{aligned}
 S &\Rightarrow (T) \Rightarrow (T, S) \Rightarrow (T, a) \Rightarrow (S, a) \Rightarrow ((T), a) \Rightarrow ((T, S), a) \Rightarrow ((T, (T)), a) \\
 &\Rightarrow ((T, (S)), a) \Rightarrow ((T, (a)), a) \Rightarrow ((T, S, (a)), a) \Rightarrow (((T, \wedge, (a)), a) \\
 &\Rightarrow ((S, \wedge, (a)), a) \Rightarrow (((T), \wedge, (a)), a) \Rightarrow (((T, S), \wedge, (a)), a) \Rightarrow \\
 &(((T, a), \wedge, (a)), a) \Rightarrow (((S, a), \wedge, (a)), a) \Rightarrow (((a, a), \wedge, (a)), a)
 \end{aligned}$$

步骤	栈	输入串	动作
0	#	$(((a, a), \wedge, (a)), a) \#$	进
1	# (	$((a, a), \wedge, (a)), a) \#$	进
2	# ((	$(a, a), \wedge, (a)), a) \#$	进
3	# (((	$a, a), \wedge, (a)), a) \#$	进
4	# (((a	$, a), \wedge, (a)), a) \#$	归约 $S \rightarrow a$.
5	# (((S	$, a), \wedge, (a)), a) \#$	归约 $T \rightarrow S$
6	# (((T	$, a), \wedge, (a)), a) \#$	进
7	# (((T,	$a), \wedge, (a)), a) \#$	进
8	# (((T, a	$), \wedge, (a)), a) \#$	归约 $S \rightarrow a$.
9	# (((T, S	$), \wedge, (a)), a) \#$	归约 $T \rightarrow T, S$
10	# (((T	$), \wedge, (a)), a) \#$	进
11	# (((T)	$, \wedge, (a)), a) \#$	归约 $S \rightarrow (T)$
12	# ((S	$, \wedge, (a)), a) \#$	归约 $T \rightarrow S$
13	# ((T	$, \wedge, (a)), a) \#$	进
14	# ((T,	$\wedge, (a)), a) \#$	进
15	# ((T, \wedge	$, (a)), a) \#$	归约 $S \rightarrow \wedge$.
16	# ((T, S	$, (a)), a) \#$	归约 $T \rightarrow T, S$

步骤	栈	输入串	动作
17.	# ((T	(a)), a) #	进
18.	# ((T,	(a)), a) #	进
19.	# ((T, (	a)), a) #	进
20.	# ((T, (a	)), a) #	归约 $S \rightarrow a$.
21.	# ((T, (S	)), a) #	归约 $T \rightarrow S$.
22.	# ((T, (T	)), a) #	进
23.	# ((T, (T)	), a) #	归约 $S \rightarrow (T)$
24.	# ((T, S	), a) #	归约 $T \rightarrow T, S$.
25.	# ((T	), a) #	进
26.	# ((T)	, a) #	归约 $S \rightarrow (T)$
27.	# (S	, a) #	归约 $T \rightarrow S$.
28.	# (T	, a) #	进
29.	# (T,	a) #	进
30.	# (T, a	) #	归约 $S \rightarrow a$.
31.	# (T, S	) #	归约 $T \rightarrow T, S$
32.	# (T	) #	进
33.	# (T)	#	归约 $S \rightarrow (T)$
34.	# S	#	接受

(3) 步骤	栈	优先关系	当前符号	剩余符号串	动作
1	#	<	(	a, (a, a) #	移进
2	# (	<	a	, (a, a) #	移进
3	# (a	>	,	(a, a) #	归约
4	# (S	<	,	(a, a) #	移进
5	# (S,	<	(	(a, a) #	移进
6	# (S, (	<	a	, a) #	移进
7	# (S, (a	>	,	a) #	归约
8	# (S, (S	<	,	a) #	移进
9	# (S, (S,	<	a	) #	移进
10	# (S, (S, a	>	)	) #	归约
11	# (S, (S, S	>	)	) #	归约
12	# (S, (T	<	)	) #	移进
13	# (S, (T)	>	)	#	归约

状态	栈	优先关系	当前符号	剩余符号串	动作
14.	#(S,S	'>	)	#	归约
15.	#(T	'<'	)	#	移进
16.	#(T)	'>	#		归约
17.	# S	'='	#		接受

3. 证明.

G_3 的所有 LR(0) 项目:

$S' \rightarrow \cdot S$ $S' \rightarrow S \cdot$

$S \rightarrow \cdot A$ $S \rightarrow A \cdot$

$A \rightarrow \cdot Ab$ $A \rightarrow A \cdot b$ $A \rightarrow Ab \cdot$ $A \rightarrow \cdot bBa$ $A \rightarrow b \cdot Ba$ $A \rightarrow bB \cdot a$ $A \rightarrow bBa \cdot$

$B \rightarrow \cdot aAc$ $B \rightarrow a \cdot Ac$ $B \rightarrow aA \cdot c$ $B \rightarrow aAc \cdot$ $B \rightarrow \cdot a$ $B \rightarrow a \cdot$ $B \rightarrow \cdot aAb$

$B \rightarrow a \cdot Ab$ $B \rightarrow aA \cdot b$ $B \rightarrow aAb \cdot$

G_3 的项集规范族:

$I_0 = \{S' \rightarrow \cdot S, S \rightarrow \cdot A, A \rightarrow \cdot Ab, A \rightarrow \cdot bBa\}$

$I_1 = \{S' \rightarrow S \cdot\}$

$I_2 = \{S \rightarrow A \cdot, A \rightarrow A \cdot b\}$

$I_3 = \{A \rightarrow b \cdot Ba, B \rightarrow \cdot aAc, B \rightarrow \cdot a, B \rightarrow \cdot aAb\}$

$I_4 = \{A \rightarrow Ab \cdot\}$

$I_5 = \{A \rightarrow bB \cdot a\}$

$I_6 = \{B \rightarrow a \cdot Ac, B \rightarrow a \cdot, B \rightarrow a \cdot Ab, A \rightarrow \cdot Ab, A \rightarrow \cdot bBa\}$

$I_7 = \{A \rightarrow bBa \cdot\}$

$I_8 = \{B \rightarrow aA \cdot c, B \rightarrow aA \cdot b, A \rightarrow A \cdot b\}$

$I_9 = \{B \rightarrow aAc \cdot\}$

$I_{10} = \{B \rightarrow aAb \cdot, A \rightarrow Ab \cdot\}$

$G_0(I_0, S) = I_1$

$G_0(I_0, A) = I_2$

$G_0(I_0, b) = I_3$

$G_0(I_3, b) = I_4$

$G_0(I_3, B) = I_5$

$G_0(I_3, a) = I_6$

$G_0(I_5, a) = I_7$

$G_0(I_6, A) = I_8$

$G_0(I_8, c) = I_9$

$G_0(I_8, b) = I_{10}$

$G_0(I_6, b) = I_3$

可以看出 I_3, I_6, I_{10} 存在冲突, 因此 G_3 不是 LR(0) 的.

对于 I_3 : $\text{FOLLOW}(S) = \{\#\}$ $\text{FOLLOW}(S) \cap \{b\} = \emptyset$ 冲突可消解

对于 I_6 : $\text{FOLLOW}(B) = \{a\}$ $\text{FOLLOW}(B) \cap \{b\} = \emptyset$ 冲突可消解.

对于 I_{10} : $\text{FOLLOW}(B) = \{a\}$ $\text{FOLLOW}(A) = \{c, b, \#\}$ $\text{FOLLOW}(A) \cap \text{FOLLOW}(B) = \emptyset$
冲突可消解

因此,所有冲突均可消解, $\therefore G_3$ 是 SLR(1) 的, 但不是 LR(0) 的

4. 解 (1) 所有 LR(0) 项目如下:

$$S' \rightarrow \cdot S \quad S' \rightarrow S \cdot$$

$$S \rightarrow \cdot AS \quad S \rightarrow A \cdot S \quad S \rightarrow AS \cdot \quad S \rightarrow \cdot b \quad S \rightarrow b \cdot$$

$$A \rightarrow \cdot SA \quad A \rightarrow S \cdot A \quad A \rightarrow SA \cdot \quad A \rightarrow \cdot a \quad A \rightarrow a \cdot$$

$$(2) I_0 = \{S' \rightarrow \cdot S, S \rightarrow \cdot AS, S \rightarrow \cdot b, A \rightarrow \cdot SA, A \rightarrow \cdot a\}$$

$$I_1 = \{S' \rightarrow S \cdot, A \rightarrow S \cdot A, A \rightarrow \cdot SA, A \rightarrow \cdot a, S \rightarrow \cdot AS, S \rightarrow \cdot b\}$$

$$I_2 = \{S \rightarrow A \cdot S, S \rightarrow \cdot AS, S \rightarrow \cdot b, A \rightarrow \cdot SA, A \rightarrow \cdot a\}$$

$$I_3 = \{S \rightarrow b \cdot\}$$

$$I_4 = \{A \rightarrow a \cdot\}$$

$$I_5 = \{A \rightarrow SA \cdot, S \rightarrow A \cdot S, S \rightarrow \cdot AS, S \rightarrow \cdot b, A \rightarrow \cdot SA, A \rightarrow \cdot a\}$$

$$I_6 = \{A \rightarrow S \cdot A, A \rightarrow \cdot SA, A \rightarrow \cdot a, S \rightarrow \cdot AS, S \rightarrow \cdot b\}$$

$$I_7 = \{S \rightarrow AS \cdot, A \rightarrow S \cdot A, A \rightarrow \cdot SA, A \rightarrow \cdot a, S \rightarrow \cdot AS, S \rightarrow \cdot b\}$$

$$G_0(I_0, S) = I_1$$

$$G_0(I_0, A) = I_2$$

$$G_0(I_0, b) = I_3$$

$$G_0(I_0, a) = I_4$$

$$G_0(I_1, A) = I_5$$

$$G_0(I_1, S) = I_6$$

$$G_0(I_1, a) = I_4$$

$$G_0(I_1, b) = I_3$$

$$G_0(I_2, A) = I_2$$

$$G_0(I_2, a) = I_4$$

$$G_0(I_2, b) = I_3$$

$$G_0(I_2, S) = I_7$$

$$G_0(I_5, S) = I_1$$

$$G_0(I_5, A) = I_2$$

$$G_0(I_5, a) = I_4$$

$$G_0(I_5, b) = I_3$$

$$G_0(I_6, S) = I_6$$

$$G_0(I_6, A) = I_5$$

$$G_0(I_6, a) = I_4$$

$$G_0(I_6, b) = I_3$$

$$G_0(I_7, S) = I_6$$

$$G_0(I_7, A) = I_5$$

$$G_0(I_7, a) = I_4$$

$$G_0(I_7, b) = I_3$$

(3) 由 (2) 可得 I_1, I_5, I_7 存在冲突

对于 I_1 : $\text{Follow}(S') = \{\#\}$

$A \rightarrow a$ $\text{Follow}(S') \cap \{a\} = \emptyset$

$S \rightarrow b$ $\text{Follow}(S') \cap \{b\} = \emptyset$ 冲突可解决

对于 I_5 : $A \rightarrow SA$ $\text{Follow}(A) = \{a, b\}$

$S \rightarrow b$ $\text{Follow}(A) \cap \{b\} \neq \emptyset$

$A \rightarrow a$ $\text{Follow}(A) \cap \{a\} \neq \emptyset$ 冲突未解决

对于 I_7 : $S \rightarrow AS$ $\text{Follow}(S) = \{\#, a, b\}$

$A \rightarrow a$ $\text{Follow}(S) \cap \{a\} \neq \emptyset$

$S \rightarrow b$ $\text{Follow}(S) \cap \{b\} \neq \emptyset$ 冲突未解决

因此 I_5, I_7 的冲突未解决

$\therefore$ 该文法 G_4 不是 SLR 文法